

Bokun Wang

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RESEARCH INTERESTS

Hardware-efficient Generative AI, Edge AI, Optimization

EDUCATION

Texas A&M University, College Station, TX

Ph.D. in Computer Science

2021-2025

Advisor: Prof. [Tianbao Yang](#)

University of Electronic Science and Technology of China, Chengdu, Sichuan, China

B.S. in Computer Science (Yingcai Honors College Program)

2014-2018

EXPERIENCE

Postdoctoral Researcher, hosted by Prof. [Diana Marculescu](#)

September 2025 -

Chandra Family Department of Electrical and Computer Engineering, The University of Texas at Austin

Machine Learning Intern

May 2023 - August 2023

Central Technology / Machine Learning Group, ARM

PROJECTS

(Ongoing) Adaptive Configuration Tuning for Hardware-Efficient Generative AI

- Improving hardware efficiency (latency, memory, energy consumption, etc.) of video and language GenAI inference while preserving generation fidelity.

📄 Manuscripts:

- ▷ Natalia Frumkin, **Bokun Wang**, et al. “DARE: Diffusion Language Model Activation Reuse for Efficient Inference.” *Under Submission*, 2026.
- ▷ Dennis Menn, Yuedong Yang, **Bokun Wang**, et al. “Training-free Latent Inter-Frame Pruning with Attention Recovery.” *Under Submission*, 2026.
- ▷ Hung-Yueh Chiang, **Bokun Wang**, Diana Marculescu. “ELANA: A Simple Energy and Latency Analyzer for LLMs.” *Technical Report*, 2026.

Data/Memory-efficient Contrastive Learning

- Developed efficient algorithms SOX, ALEXR, and SCENT with provable convergence guarantees, adopted by open-source projects including [LibAUC](#), [FastCLIP](#), and [DRRho-CLIP](#).
- Proposed an algorithm based on discriminative probabilistic modeling and Monte Carlo integration for bi-modal self-supervised representation learning, offering improved generalization guarantees. It achieves superior overall performance across 8 downstream retrieval and zero-shot classification tasks compared to 5 representative baselines. Notably, it outperforms the widely used CLIP algorithm by over 15%.
- Implemented the project using Python and PyTorch, and open-sourced the code on [GitHub](#).

■ Publications:

- ▷ Xiyuan Wei, Linli Zhou, **Bokun Wang**, et al. “A Geometry-Aware Efficient Algorithm for Compositional Entropic Risk Minimization.” In *International Conference on Machine Learning (ICML)*, 2026.
- ▷ Vicente Balmaseda, **Bokun Wang**, et al. “Discovering Global False Negatives On the Fly for Self-supervised Contrastive Learning.” In *International Conference on Machine Learning (ICML)*, 2025.
- ▷ **Bokun Wang**, et al. “A Near-Optimal Single-Loop Algorithm for Convex Finite-Sum Coupled Compositional Stochastic Optimization.” In *International Conference on Machine Learning (ICML)*, 2025.
- ▷ **Bokun Wang**, et al. “On Discriminative Probabilistic Modeling for Self-Supervised Representation Learning.” In *International Conference on Learning Representations (ICLR)*, 2025;
- ▷ **Bokun Wang**, et al. “Finite-Sum Coupled Compositional Stochastic Optimization Theory and Applications.” In *International Conference on Machine Learning (ICML)*, 2022;

Efficient Imbalanced and Multi-instance Medical Image Analysis

- Designed efficient, provably convergent algorithms for (partial) AUC maximization, tailored to the challenges of imbalanced medical datasets and high-resolution imaging data such as CT scans.
- Our proposed algorithms are GPU memory-efficient, unlike prior methods whose performance is limited by memory constraints (For example, they cannot handle large histopathological images).
- Demonstrated the effectiveness and superiority of our algorithms on benchmark datasets and real-world high-resolution medical image datasets.

■ Publications:

- ▷ Dixian Zhu, **Bokun Wang**, et al. “Provable Multi-instance Deep AUC Maximization with Stochastic Pooling.” In *International Conference on Machine Learning (ICML)*, 2023.
- ▷ Dixian Zhu, Gang Li, **Bokun Wang**, et al. “When AUC meets DRO: Optimizing Partial AUC for Deep Learning with Non-Convex Convergence Guarantee.” In *International Conference on Machine Learning (ICML)*, 2022.

HONORS & AWARDS

Selected Participant for FutureBAProf, Tippie College of Business, The University of Iowa, 2026

Gold Reviewer Award, International Conference on Machine Learning (ICML), 2026

Notable Reviewer, International Conference on Learning Representations (ICLR), 2025

Oral Presentation, Conference on the Mathematical Theory of Deep Neural Networks (DeepMath), 2024

Scholar Award, Conference on Neural Information Processing Systems (NeurIPS), 2022

Travel Award, International Conference on Machine Learning (ICML), 2022

Spotlight Presentation, International Conference on Learning Representations (ICLR), 2022

Provincial Outstanding Graduate, Sichuan Province, China, 2018

Lixin Tang Scholarship, University of Electronic Science and Technology of China, 2017

National Scholarship, Ministry of Education of the People's Republic of China, 2016

People's Scholarship (Special Class), University of Electronic Science and Technology of China (UESTC), 2015

Freshman Scholarship, University of Electronic Science and Technology of China (UESTC), 2014

SKILLS

Languages and Libraries: Python, PyTorch, Jax, Scikit-learn, NumPy, Pandas, Matplotlib

Tools: Git, Docker, Slurm, LaTeX